

Shenghan Gao

PHOTONIC INTEGRATED CIRCUITS · PHOTONICS HARDWARE · VALIDATION & TEST

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Summary

Ph.D. candidate in Electrical and Computer Engineering specializing in integrated photonics, photonic hardware validation, and coherent electro-optic systems. End-to-end experience across silicon photonics and TFLN/LNOI PIC design, passive and active photonic devices, cleanroom fabrication, fiber-to-chip integration, RF/EO characterization, DSP-based signal analysis, and Python/PyVISA/SCPI automated measurement. Seeking PIC design, silicon photonics, photonics validation/test, electro-optic device, and optical hardware R&D roles.

Core Skills

PIC design	Silicon photonics and TFLN/LNOI PIC design; waveguides, directional couplers, MMI/splitters, grating/edge couplers, interferometers, electro-optic modulators, thermal phase shifters
Simulation/layout	GDSFactory, Cadence Virtuoso, KLayout; Lumerical, RSoft, COMSOL, Zemax/CodeV; optical mode analysis, device optimization, PIC layout preparation
Fab/integration	Cleanroom fabrication, spin coating, photoresist processing, electron-beam lithography, etching, SEM inspection; fiber-array coupling, fiber-to-chip coupling, photonic packaging, optical alignment
Validation	Device characterization, high-speed EO/RF characterization, VNA, RF detection, high-speed/balanced photodetection, EOMs, OSAs, SVAs, spectrum analyzers, oscilloscopes, tunable lasers
Automation/DSP	Python measurement automation, PyVISA/SCPI, NI MAX, automated testing and data acquisition, parameter sweeps, data logging, FFT/frequency-domain analysis, DSP equalization, MATLAB
Coherent optics	Optical injection locking, semiconductor-laser rate equations, coherent interference, carrier recovery, AM/PM modulation transfer, PID phase stabilization

Research and Education Experience

University of Pittsburgh

Pittsburgh, PA

PH.D. IN ELECTRICAL AND COMPUTER ENGINEERING, EXPECTED DEC. 2026 | RESEARCH ASSISTANT | ADVISOR: NATHAN

Spring 2020 - present

YOUNGBLOOD

- Developed an optical-injection-locking framework for distributed coherent photonic computing, combining semiconductor-laser rate-equation modeling, injection-ratio/detuning sweeps, coherent carrier recovery, and balanced coherent detection.
- Designed chip-scale photonic integrated circuits to translate fiber injection-locking systems onto integrated platforms, including waveguides, directional couplers, MMI/splitters, grating couplers, edge-coupling interfaces, and interferometric paths.
- Simulated silicon photonics and TFLN/LNOI passive and active photonic devices using GDSFactory, Cadence Virtuoso, KLayout, Lumerical, RSoft, and COMSOL workflows for layout, optical mode analysis, electro-optic modulation, and device optimization.
- Performed and supported photonic cleanroom fabrication and process evaluation, including spin coating/photoresist processing, e-beam lithography, etching, SEM inspection, and characterization of passive and active photonic structures.
- Integrated and aligned photonic chips using fiber arrays, grating-coupler alignment, edge coupling, optical packaging, VOAs, polarization controllers, and tunable-laser setups to connect fabricated devices to validation testbeds.
- Built high-speed EO/RF validation workflows using EOMs, high-speed photodetectors, balanced photodetection, OSAs, SVAs, spectrum analyzers, VNA-based characterization, RF detection, lock-in detection, PID phase feedback, and fiber-stretcher control.
- Automated photonic hardware validation with Python, PyVISA/SCPI, and NI MAX for wavelength, power, injection-ratio, and frequency sweeps; logged data and generated FFT/DSP analysis, plots, and repeatable reports.

Georgia Institute of Technology

Atlanta, GA

M.S. IN COMPUTER SCIENCE, MACHINE LEARNING TRACK, EXPECTED MAY 2027

Spring 2024 - present

- Completed graduate CS coursework with A grades in Machine Learning, Deep Learning, Computer Vision, Machine Learning for Trading, Game AI, AI Ethics and Society, and Introduction to Cognitive Science.
- Built machine-learning project experience around supervised learning, unsupervised learning, reinforcement learning, randomized optimization, model selection, cross-validation, and error analysis; project datasets and metrics to be specified [details TBD].
- Developed deep-learning and computer-vision project skills in neural-network training, backpropagation, gradient-based optimization, CNN/RNN architectures, feature extraction, image geometry, and model evaluation for perception-oriented tasks [details TBD].
- Applied Python-based ML workflows to trading and decision-making experiments, including regression, KNN/regression-tree learners, Q-learning, data preprocessing, parameter sweeps, and performance visualization.

University of Rochester

Rochester, NY

M.S. IN OPTICS, AUG. 2019 | B.S. IN OPTICAL ENGINEERING, MAY 2017 | B.S. IN APPLIED MATHEMATICS, MAY 2017

Sep 2017 - Aug 2019

- Research Assistant in THz photonics under Xi-Cheng Zhang; generated THz signals from liquid plasmas and quantified how temperature, polarity, and viscosity affected emission strength, bandwidth, repeatability, and spectral content.
- Processed time- and frequency-domain experimental data using SNR, sensitivity, regression, and repeatability analysis; prototyped digital cloaking with lenslet arrays and MATLAB pixel remapping.

Industry Experience

Lumentum Operations LLC

San Jose, CA

RESEARCH INTERN | MENTOR: JOE WEN

May 2023 - Aug 2023

- Developed MATLAB/Python FIR-filter equalization workflows for communication signals distorted by intersymbol interference, connecting DSP filter design to recovered waveform quality.
- Collected measured signal data and applied FFT/frequency-domain analysis to compare filter responses, tune coefficients, and validate demodulated-signal recovery across test cases.

Sunny Optics

Ningbo, China

RESEARCH INTERN | MENTOR: CHAO XU

May 2018 - Aug 2018

- Designed and optimized automotive LiDAR and mobile-camera lens systems in CodeV/Zemax against optical performance, tolerance, and packaging requirements.
- Collaborated with opto-mechanical engineers during lab integration, using alignment and test feedback to refine optical system functionality.

Selected Publications and Talks

MOSAIC: A Multicore Optical Architecture for Scaling Up AI Acceleration

L. LIU, R. ZHU, **S. GAO**, X. XIN, Y. ZHANG, N. YOUNGBLOOD, J. YANG, SUBMITTED TO HPCA 2027, UNDER REVIEW, 2026

Distributed Coherent Optical Computing via Injection-Locked Photonic Networks

GAO, S., K. LÜDGE, F. DA ROS, N. YOUNGBLOOD, SUBMITTED TO APL, UNDER REVIEW, 2026

Distributed Coherent Photonic Computing via Injection Locking

GAO, S., SPIE OPTICS + PHOTONICS (ORAL PRESENTATION), 2026

Distributed Coherent Optical Computing via Injection-Locked Photonic Networks

GAO, S., INVITED PHD CANDIDATE TALK, PITTSBURGH QUANTUM INSTITUTE, PITTSBURGH, 2025

Strain-free FBG sensors for accurate temperature measurements from cryogenic temperatures to 300 K

GAO, S., P. R. SWINEHART, S. ZHONG, J. WUENSCHALL, N. LALAM, M. BURIC, K. P. CHEN, SPIE DEFENSE + COMMERCIAL SENSING (ORAL PRESENTATION), 2023

Preference of subpicosecond laser pulses for terahertz wave generation from liquids

Q. JIN, Y. E., **S. GAO**, X.-C. ZHANG, *Advanced Photonics* 2(1), 015001, 2020

Investigation of liquid properties on emitting terahertz wave under ultrashort optical excitation

S. GAO, Q. JIN, Y. E., X.-C. ZHANG, SPIE, 2019